PROJECT REPORT

SONAR ROCK VS MINE PREDICTION USING MACHINE LEARNING

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Course: Fundamentals of AI and ML

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Introduction :-

Machine learning is a branch of artificial intelligence that enables systems to learn from data and improve their performance without being explicitly programmed. It is widely used in various real world applications such as healthcare, finance, and defense.

In this project, a machine learning model is developed to classify sonar signals as either rock or mine. Sonar systems are used to detect objects underwater by sending sound waves and analyzing the reflected signals. The classification of such signals is important in naval operations and underwater exploration

Objective

The dataset used in this project is the sonar dataset obtained from the UCI Machine Learning Repository. It contains:

- **60 numerical features**
- **Each feature represents the energy of sonar signals at different frequencies**
- **A target label that indicates the class**

<u>The output label has two values:</u>

R which represents Rock

M which represents Mine

The dataset is commonly used for binary classification problems.

Technologies Used:-

The following technologies and libraries are used in this project:

- **Python programming language**
- **NumPy for numerical operations**
- **Pandas for data handling and analysis**
- **Scikit learn for machine learning algorithms**

Methodology :-

- **Data Collection**

The dataset is loaded using the pandas library from a CSV file. The data is stored in a structured format for further processing.

- **Data Analysis**

Basic analysis is performed to understand the dataset. Functions like shape, describe, and value counts are used to examine the data distribution and statistical properties.

- **Data Preprocessing**

The dataset is divided into input features and output labels.

Input features X contain the 60 numerical values

Output labels Y contain the classification values R or M

- **Train Test Split**

The dataset is divided into training and testing sets.

Training data is used to train the model

Testing data is used to evaluate the model

A stratified split is used to maintain equal distribution of classes in both sets.

- **Model Training**

Logistic Regression algorithm is used for training the model. It is suitable for binary classification problems and provides good performance on structured datasets.

- **Model Evaluation**

The performance of the model is evaluated using accuracy score.

Training accuracy measures how well the model fits the training data

Testing accuracy measures how well the model performs on unseen data

- **Prediction System**

A predictive system is developed where new sonar input data is provided to the model. The model processes the input and predicts whether the object is a rock or a mine.

Results :-

The model achieves good accuracy on both training and testing datasets. It successfully classifies sonar signals into rock and mine categories. The results indicate that logistic regression is effective for this type of classification problem.

Conclusion :-

This project demonstrates the application of machine learning in solving classification problems. Logistic regression proves to be a simple and efficient algorithm for binary classification tasks. The model can be used in real world scenarios such as underwater object detection.

References

- Scikit learn documentation
- NumPy documentation
- Pandas documentation
- UCI Machine Learning Repository